

4.13.46

EE25BTECH11042 - Nipun Dasari

Question:

A straight line L is perpendicular to the line $5x - y + 1 = 0$. The area of the triangle formed by L and the coordinate axes is 5. Find the equation of the line L .

Solution:

Let the given line be L_1 and the required perpendicular line be L . The equation of the given line L_1 is $5x - y = -1$. Using the matrix form $\mathbf{n}^T \mathbf{x} = c$, its parameters are:

$$\mathbf{n}_1 = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c_1 = -1 \quad (0.1)$$

Let the required line L have the equation $\mathbf{n}_2^T \mathbf{x} = c_2$. Since the lines are perpendicular, the dot product of their normal vectors must be zero, so $\mathbf{n}_1^T \mathbf{n}_2 = 0$.

Let's assume the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ m \end{pmatrix}$. We can solve for m :

$$\begin{aligned} \mathbf{n}_1^T \mathbf{n}_2 &= \begin{pmatrix} 5 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ m \end{pmatrix} = 5 - m = 0 \\ \implies m &= 5 \end{aligned} \quad (0.2)$$

Therefore, the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, and its equation is $x + 5y = c_2$.

The line L forms a triangle with the coordinate axes. The equation for the y -axis is $x = 0$, which is $\mathbf{e}_1^T \mathbf{x} = 0$ with $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. The equation for the x -axis is $y = 0$, which is $\mathbf{e}_2^T \mathbf{x} = 0$ with $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

The x -intercept, $\mathbf{a}$, is the intersection of line L and the x -axis. We solve the stacked system:

$$\begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix} \quad (0.3)$$

The intercept vector $\mathbf{a}$ is found by inverting the matrix:

$$\mathbf{a} = \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix} \quad (0.4)$$

The y -intercept, $\mathbf{b}$, is the intersection of line L and the y -axis:

$$\begin{pmatrix} 1 & 5 \\ 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix} \quad (0.5)$$

The intercept vector $\mathbf{b}$ is:

$$\mathbf{b} = \begin{pmatrix} 1 & 5 \\ 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \frac{1}{-5} \begin{pmatrix} 0 & -5 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ c_2/5 \end{pmatrix} \quad (0.6)$$

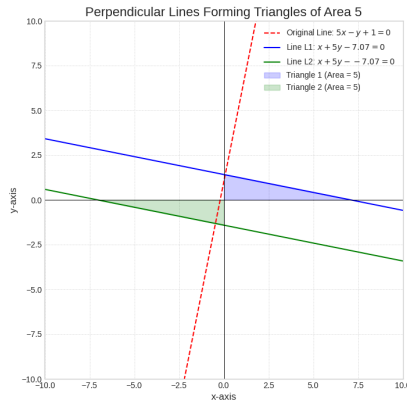
The area of the triangle formed by the origin, $\mathbf{a}$, and $\mathbf{b}$ is 5. Since $\mathbf{a}$ and $\mathbf{b}$ lie on the axes, the angle θ between them is 90° .

$$\begin{aligned} \text{Area} &= \frac{1}{2} \|\mathbf{a}\| \|\mathbf{b}\| \sin(90^\circ) = 5 \\ \frac{1}{2} \left(\sqrt{c_2^2 + 0^2} \right) \left(\sqrt{0^2 + (c_2/5)^2} \right) (1) &= 5 \\ \frac{1}{2} |c_2| \left| \frac{c_2}{5} \right| &= 5 \\ \frac{|c_2^2|}{10} &= 5 \\ c_2^2 = 50 &\implies c_2 = \pm \sqrt{50} = \pm 5\sqrt{2} \end{aligned} \quad (0.7)$$

There are two possible lines that satisfy the given conditions. Substituting the values of c_2 into the matrix equation $\mathbf{n}_2^\top \mathbf{x} = c_2$, the possible equations for line L are:

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5\sqrt{2} \quad (0.8)$$

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -5\sqrt{2} \quad (0.9)$$



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